

CLUSTER STRUCTURE OF SCHUBERT CELL

1. PRELIMINARIES

The purpose of this note is to construct a cluster structure over Schubert cell BwB/B . We will restrict to type A since the diagram is easier to visualize.

The reference is [1], [2] and [3]. They dealt with double Bruhat cell, corresponding to the case $BwB \cap B^- = BwB \cap B^-eB^-$.

1.1. **Pin.** Let G be a reductive group. For each simple root α_i , we have a group homomorphism

$$SL_2 = SL_2(\alpha_i) \longrightarrow G.$$

We will denote

$$\begin{aligned} x_i : \mathbb{C} &\longrightarrow G, & x_i(m) &= \text{image of } \begin{bmatrix} 1 & m \\ 0 & 1 \end{bmatrix} \\ y_i : \mathbb{C} &\longrightarrow G, & y_i(t) &= \text{image of } \begin{bmatrix} 1 & 0 \\ t & 1 \end{bmatrix} \\ \chi_i : \mathbb{C} &\longrightarrow G, & \chi_i(z) &= \text{image of } \begin{bmatrix} z & 0 \\ 0 & z^{-1} \end{bmatrix} \end{aligned}$$

$$\dot{s}_i = \text{image of } \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \ddot{s}_i = \text{image of } \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \dot{s}_i^{-1}.$$

Since $\dot{s}_i$ and $\ddot{s}_i$ satisfy braid relations, we can define $\dot{w}$ and $\ddot{w}$ for any $w \in W$.

1.2. **Minors.** For a dominant weight λ and $w \in W$, we define

$$\eta_{w\lambda} = \dot{w}\eta_\lambda, \quad \eta_\lambda = \text{the highest weight vector in } V^\lambda.$$

We define the generalized minor to be

$$\begin{aligned} \Delta_{w\lambda}^{u\lambda}(g) &= \text{coefficient of } \eta_{u\lambda} \text{ in } g\eta_{w\lambda} \\ &= \text{coefficient of } \eta_\lambda \text{ in } \dot{u}^{-1}g\dot{w}\eta_\lambda \\ &= \Delta_\lambda^\lambda(\dot{u}^{-1}g\dot{w}) = \Delta_\lambda^\lambda(\ddot{v}g\dot{w}). \end{aligned}$$

Example. Let $\lambda = \varpi_k = \epsilon_1 + \cdots + \epsilon_k$ be the k -th fundamental weight. Then $V^\lambda = \Lambda^k \mathbb{C}^n$. For a subset $A = \{a_1 < \cdots < a_k\}$ of $[n]$, we denote

$$\epsilon_A := \epsilon_{a_1} + \cdots + \epsilon_{a_k}, \quad e_A = e_{a_1} \wedge \cdots \wedge e_{a_k}.$$

Then

$$\Delta_{w\varpi_k}^{u\varpi_k}(g) = \text{coefficient of } e_{u[k]} \text{ in } ge_{w[k]} = \Delta_{u[k]}^{w[k]}(g)$$

where

$$\Delta_A^B(g) = \det(g_{ij})_{i \in A, j \in B}.$$

Precisely, assume $B = \{b_1 < \cdots < b_k\}$, then

$$\begin{aligned} ge_B &= ge_{b_1} \wedge \cdots \wedge ge_{b_k} \\ &= \left(\sum_{a_1} g_{a_1 b_1} e_{a_1} \right) \wedge \cdots \wedge \left(\sum_{a_k} g_{a_k b_k} e_{a_k} \right) \\ &= \sum_{a_1, \dots, a_k} g_{a_1 b_1} \cdots g_{a_k b_k} e_{a_1} \wedge \cdots \wedge e_{a_k} = \sum_A \Delta_A^B(g) e_A. \end{aligned}$$

2. THE SEED

2.1. The quiver. Let us pick a reduced word of w

$$w = s_{i_1} \cdots s_{i_\ell}.$$

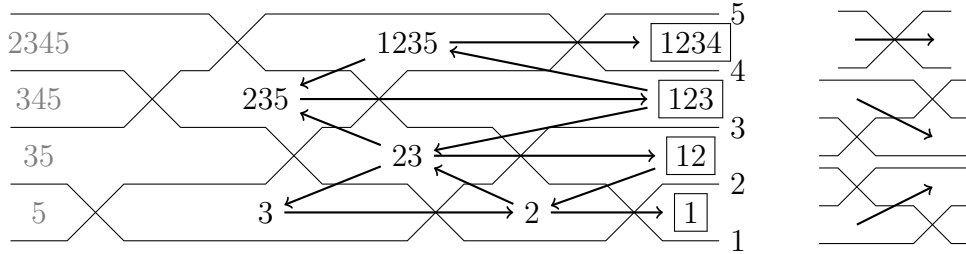
Let us define a quiver with vertices $\{1, \dots, \ell\}$. For $1 \leq k \leq \ell$, we define

$$k^+ = \min(r > k : i_r = i_k) \in \{1, \dots, \ell\} \cup \{\infty\}.$$

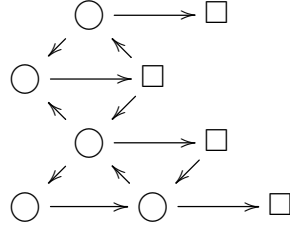
We frozen those k with $k^+ = \infty$. We construct a quiver by

$$k \rightarrow m \iff \begin{cases} k < m = k^+, & i_k = i_m, \\ m < k < m^+ < k^+, & i_k \neq i_m. \end{cases}$$

In type A , we can visualize the reduced word by wiring diagram.



The quiver is



2.2. The cluster variables. For each $k \in \{1, \dots, \ell\}$, we define a minor

$$M_k = \Delta_{\varpi_{i_k}}^{w_{>k}^{-1} \varpi_{i_k}}, \quad \text{where } w_{>k}^{-1} = s_{i_\ell} \cdots s_{i_{k+1}}.$$

The function M_k is a function over G , there are two ways of viewing it as a function over $Bw^{-1}B/B$. For example, in the above example, each chamber is labeled by a subset A if $w_{>k}^{-1} \varpi_{i_k} = \epsilon_A$. For a subset $A \subset [n]$ of size k corresponds to the minor $\Delta_A^{[k]}$.

- We think

$$M_k(gB) = \frac{\Delta_{\varpi_{i_k}}^{w_{>k}^{-1} \varpi_{i_k}}(g)}{\Delta_{\varpi_{i_k}}^{w^{-1} \varpi_{i_k}}(g)}, \quad gB \in Bw^{-1}B/B.$$

This does not depend on the choice of representative g . Note that any element $g \in Bw^{-1}B$, the denominator

$$\Delta_{\varpi_{i_k}}^{w^{-1} \varpi_{i_k}}(g) \neq 0.$$

- We can lift (recall that $\ddot{w}^{-1} = \dot{m}$ for $m = w^{-1}$)

$$G \supset U\ddot{w}^{-1} \cap \ddot{w}^{-1}U^- \xrightarrow{\sim} Bw^{-1}B/B, \quad g \longmapsto gB/B.$$

Then we can restrict M_k from G to $U\ddot{w}^{-1} \cap \ddot{w}^{-1}U^-$ and pass to $Bw^{-1}B/B$ by the isomorphism. Note that under this parametrization, we have $\Delta_{\varpi_{i_k}}^{w^{-1} \varpi_{i_k}}(g) = 1$.

We will show the seed gives an upper cluster structure over

$$\mathbb{C}[\mathring{R}_{e,w^{-1}}].$$

Here we invert the frozen variables.

3. STARFISH LEMMA

Starfish Lemma. Let R be a $\mathbb{C}$ -algebra and a normal Noetherian domain. Assume we are given a quiver Q and cluster variables $\{A_i\}_{i \in Q_0} \subseteq R$ such that

- (1) $\{A_i\}_{i \in Q_0}$ forms a transcendental basis of $\text{Frac}(R)$;
- (2) $\{A_i\}_{i \in Q_0^{\text{nf}}}$ are pairwise coprime;
- (3) $\mu_k(A_k) \in R$ and it coprime to A_k .

Then R contains the upper cluster algebra U of Q .

The property (1) shows that R and U has the same fractional field. Let $f \in U$. We can assume $f = \frac{g}{h}$ for coprime $g, h \in R$. We have

$$f = \frac{\text{a polynomial in } \{A_i\}_{i \in Q_0}}{\text{a monomial in } \{A_i\}_{i \in Q_0^{\text{uf}}}} \implies h \mid \text{the denominator.}$$

On the other hand, f also admits the same expression with $\{A_i\}$ replaced by $\{\mu_k(A_i)\} = \{A_r : r \neq k\} \cup \{\mu_k(A_k)\}$. Thus $h_1 \mid \mu_k(A_r)$ for some r . This leads to a contradiction.

$$\epsilon_{ij} = \#\{i \rightarrow j\} - \#\{j \rightarrow i\}$$
$$\mathcal{U} = \mathbb{C}[A_i^\pm] \cap \bigcap_{k \in Q_0^{\text{uf}}} \mathbb{C}[\mu_k(A_i)^\pm].$$

Proof. We notice that if $k < m$, then there are no arrows between k and m^+ . If $k = m$, we have an arrow from k to m^+ . This proves the submatrix $(\epsilon_{m^+, k})_{k, m \in Q_0^{\text{uf}}}$ is unipotent. $\square$

	1	2	3	4	6
$1^+ = 6$	-1	0	0	1	0
$2^+ = \boxed{5}$	0	-1	1	1	0
$3^+ = \boxed{8}$	0	0	-1	0	0
$4^+ = \boxed{7}$	0	0	0	-1	1
$6^+ = \boxed{9}$	0	0	0	0	-1

Lemma. Let X be a normal affine variety. Let Q be a quiver and $\{A_i\}_{i \in Q_0} \subseteq \mathbb{C}[X]$ with full-rank condition above. Assume

(1) we have an isomorphism

$$U := \{x \in X : A_i(x) \neq 0\} \xrightarrow{\sim} (\mathbb{C}^\times)^{Q_0}, \quad x \mapsto (A_i(x))$$

(2) for each $k \in Q_0^{\text{uf}}$, $\mu_k(A_k) \in \mathbb{C}[X]$ is still regular over X , and

(3) we have an isomorphism

$$U_k := \{x \in X : \mu_k(A_i)(x) \neq 0\} \xrightarrow{\sim} (\mathbb{C}^\times)^{Q_0}, \quad x \mapsto (\mu_k(A_i)(x)).$$

Then $\mathbb{C}[X]$ coincides with the cluster algebra $\mathcal{U}$ of Q .

Proof. Note that

$$\mathbb{C}[U] = \mathbb{C}[A_i^{\pm 1}], \quad \mathbb{C}[U_k] = \mathbb{C}[\mu_k(A_i)^{\pm 1}].$$

By Lemma above,

$$\mathcal{U} = \left\{ f \in \text{Frac } \mathbb{C}[X] : \begin{array}{l} f|_U \text{ is regular} \\ f|_{U_k} \text{ is regular} \end{array} \right\}.$$

Obviously $\mathbb{C}[X] \subseteq \mathcal{U}$.

Let us first prove the inclusion $\mathcal{U} \subseteq \mathbb{C}[X]$ using starfish Lemma. Since U is an open dense subset of X , the property (1) verifies (1).

To see (2) and (3), we need the following Lemma.

Lemma. A Laurent monomial in (unfrozen) variables $\{A_i\}$ is regular if and only if it is a usual monomial.

Proof. Assume f is such a monomial. If the exponent of A_k is $-m < 0$, then

$$f = \mu_k(A_k)^m \cdot \frac{\text{a monomial in } \{A_r\}_{r \neq k}}{(\text{a binomial in } \{A_r\}_{r \neq k})^m}$$

which has a pole over U_k (recall that $\mu_k(A_k) \neq 0$ over U_k) and there exists some point such that the denominator is 0. This leads to a contradiction. $\square$

Note that A_k is actually irreducible. Assume $P \mid A_k$, by restricting to U , we see P is invertible over U , i.e. a Laurent monomial in $\{A_k\}$ up to a constant. By multiplying a unit, we can assume P is a Laurent monomial in $\{A_k\}_{k \in Q_0^{\text{uf}}}$. By Lemma above, they are actually usual monomial. This forces $P = A_k$ or $P = 1$.

If $P \mid A_k$ and $P \mid A_r$, we can similarly assume P is a monomial, and this force $P = 1$. This proves property (2).

Since A_k is irreducible, it suffices to show $A_k \nmid \mu_k(A_k)$. Since over U , the variable

$$\mu_k(A_k) = \frac{\text{a binomial of } \{A_r\}_{r \neq k}}{A_k}$$

has an zero, while $A_k \neq 0$. This proves (3). $\square$

Remark. Actually, we can show $\mathbb{C}[\mathring{R}_{e,w^{-1}}]$ is a factorial ring. Since it is obtained by localization of coordinate ring of $Bw^{-1}B/B \cong \mathbb{C}^{\ell(w)}$.

4. PROOF OF (2)

In this section, we will show $M'_k := \mu_k(M_k)$ is regular.

4.1. Relations.

Fundamental relation [2, Theorem 1.17]. If $us_i > u$ and $vs_i > v$, then

$$\Delta_{u\varpi_i}^{v\varpi_i} \Delta_{us_i\varpi_i}^{vs_i\varpi_i} - \Delta_{u\varpi_i}^{vs_i\varpi_i} \Delta_{us_i\varpi_i}^{v\varpi_i} = \prod_{j \rightarrow i} \Delta_{u\varpi_j}^{v\varpi_j} = \prod_{j \neq i} (\Delta_{u\varpi_j}^{v\varpi_j})^{-\langle \alpha_j^\vee, \alpha_i \rangle}.$$

Proof. It suffices to show when $u = v = \text{id}$. When restricting to torus, it gives

$$t^{\varpi_i} t^{s_i \varpi_i} - 0 = t^{-\sum \langle \alpha_j^\vee, \alpha_i \rangle \varpi_j}.$$

We will view both sides as elements of $\mathbb{C}[G]$, equipped with an action of $G \times G$ by left and right multiplication. Both sides are of the same weight,

$$\text{LHS}(t_1 g t_2) = (t_1 t_2)^{\varpi_i + s_i \varpi_i} \text{LHS}(g)$$

$$\text{RHS}(t_1 g t_2) = (t_1 t_2)^{-\sum \langle \alpha_j^\vee, \alpha_i \rangle \varpi_j} \text{RHS}(g).$$

Since

$$\mathbb{C}[G] = \bigoplus_{\lambda \text{ dominant}} (V^\lambda)^* \boxtimes V^\lambda$$

with each $G \times G$ -highest weight appears of multiplicity one. It suffices to show both sides are highest weight vector. This is clear for the RHS. We need to check

$$\text{LHS}(g x_j(t)) = \text{LHS}(g), \quad \text{LHS}(x_j(t) g) = \text{LHS}(g).$$

This is obvious for $j \neq i$ as

$$x_j(t) \eta_{\varpi_i} = \eta_{\varpi_i}, \quad x_j(t) \eta_{s_i \varpi_i} = \eta_{s_i \varpi_i}$$

as both $x_j(t)$ and $\dot{s}_i^{-1} x_j(t) \dot{s}_i$ lie in U . For $i = j$, it reduces to the computation of SL_2 . Assume

$$g \begin{bmatrix} \eta_{\varpi_i} & \eta_{s_i \varpi_i} \end{bmatrix} = \begin{bmatrix} \eta_{\varpi_i} & \eta_{s_i \varpi_i} \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \text{other weight space}.$$

Then

$$\text{LHS}(g) = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Since $x_j(t)$ preserves $\text{span}(\eta_{\varpi_i}, \eta_{s_i \varpi_i})$ and other weight spaces, we have

$$\text{LHS}(g x_i(t)) = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} = \text{LHS}(g),$$

$$\text{LHS}(x_i(t) g) = \det \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{LHS}(g).$$

We are done. $\square$

Example. When $u = v = e$ and $i = 1$ in SL_n , this tells that

$$\Delta_1^1 \Delta_2^2 - \Delta_1^2 \Delta_2^1 = \Delta_{12}^{12}.$$

When $u = v = e$ and $i = 2$, this tells that

$$\Delta_{12}^{12} \Delta_{13}^{13} - \Delta_{12}^{13} \Delta_{13}^{12} = \Delta_1^1 \Delta_{123}^{123}.$$

It suffices to show for

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}.$$

By multiplying a scalar matrix, we can assume $\det X = 1$. Recall the operator

$$X \mapsto X^* := ((-1)^{i+j} \Delta_{[n] \setminus \{i\}}^{[n] \setminus \{j\}}).$$

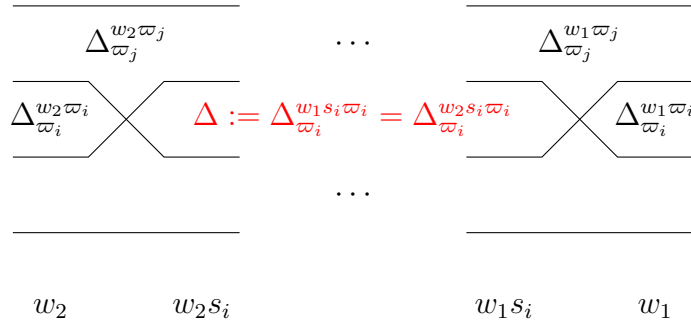
Since $\det X = 1$, we have $X^* = X^{-1}$, and we thus have $X = (X^*)^*$. Then

$$\Delta_1^1 \Delta_{123}^{123} = x_{11} = \text{the } 23\text{-}23 \text{ minor of } X^* = \begin{bmatrix} \Delta_{23}^{23} & -\Delta_{23}^{13} & \Delta_{23}^{12} \\ -\Delta_{13}^{23} & \Delta_{13}^{13} & -\Delta_{13}^{12} \\ \Delta_{12}^{23} & -\Delta_{12}^{13} & \Delta_{12}^{12} \end{bmatrix} = \Delta_{13}^{13} \Delta_{12}^{12} - \Delta_{12}^{13} \Delta_{13}^{12}.$$

Corollary. If $us_i < s_i$ and $vs_i > v$, then the above relation should be modified to $-\text{LHS} = \text{RHS}$.

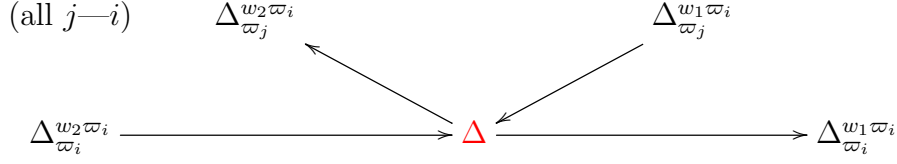
Proof. This is by replacing u by us_i . Note that $us_i \varpi_j = u \varpi_j$ if $j \neq i$. $\square$

4.2. Proof of (2). Now let us prove (2). We actually can show all M'_k is regular over G . Consider a closed chamber at the i -th level. Consider the left end and the right end.



Let Δ' be the mutation of the variable Δ . Let us first prove when each $j-i$ appears in the interval $i_k, \dots, i_{k+}$, i.e. the minors $\Delta_{w_2 s_i}^{w_2 s_i}$ and $\Delta_{w_1 s_i}^{w_1 s_i}$ appear in two different chambers.

The quiver is



Then we have

$$\begin{aligned}
\Delta \Delta' &= \Delta_{w_i}^{w_1 w_i} \prod_{j \text{---} i} \Delta_{w_j}^{w_2 w_i} + \Delta_{w_i}^{w_2 w_i} \prod_{j \text{---} i} \Delta_{w_j}^{w_1 w_i} \\
&= \Delta_{w_i}^{w_1 w_i} (\Delta_{w_i}^{w_2 w_i} \Delta_{s_i w_i}^{w_2 s_i w_i} - \Delta_{s_i w_i}^{w_2 w_i} \Delta_{w_i}^{w_2 s_i w_i}) \\
&\quad + \Delta_{w_i}^{w_2 w_i} (\Delta_{s_i w_i}^{w_1 w_i} \Delta_{w_i}^{w_1 s_i w_i} - \Delta_{w_i}^{w_1 w_i} \Delta_{s_i w_i}^{w_1 s_i w_i}) \\
&= \Delta (\Delta_{w_i}^{w_2 w_i} \Delta_{s_i w_i}^{w_1 w_i} - \Delta_{w_i}^{w_1 w_i} \Delta_{s_i w_i}^{w_2 w_i}).
\end{aligned}$$

Thus the mutation Δ' is regular over G .

In general, there could be some $j \text{---} i$ not appearing in $i_k, \dots, i_{k+}$, i.e. $\Delta_{w_j}^{w_1 w_i}$ and $\Delta_{w_j}^{w_2 w_i}$ appear in the same chamber. For such j , we have

$$\Delta_{w_j}^{w_1 w_i} = \Delta_{w_j}^{w_2 w_i} = \Delta_{w_j}^{w_1 s_i w_i} = \Delta_{w_j}^{w_2 s_i w_i}.$$

Then in general we have

$$\Delta \Delta' \cdot \prod_{j \text{ not appear}} \Delta_{w_j}^{w_1 w_i} = \text{the RHS above.}$$

Then the RHS is divided by $\Delta = \Delta_{w_i}^{w_1 s_i w_i}$ and $\Delta_{w_j}^{w_1 s_i w_i}$ for j not appear. Actually the RHS is divided by the product of them by the following Lemma. That is, Δ' is regular.

Lemma. For $i \neq j$, $\Delta_{w w_i}^{u w_i}$ and $\Delta_{w w_j}^{u w_j}$ are coprime.

Proof. By replacing g by $\dot{w} g \dot{u}^{-1}$, it suffices to show when $u = w = \text{id}$. This follows from the fact that their zero loci are distinct opposite Schubert divisors $\overline{B^- s_i B}$. $\square$

5. PROOF OF (1)

To prove (1), there are two steps.

- (Step 1) We will use a parametrization of BwB/B ;
- (Step 2) Define a twist map $BwB/B \cong Bw^{-1}B/B$.

5.1. **Step 1.** Recall we fixed a reduced word

$$w = s_{i_1} s_{i_2} \cdots s_{i_\ell}.$$

Lemma. We have a birational map

$$(\mathbb{C}^\times)^\ell \longrightarrow BwB/B \quad (t_1, \dots, t_\ell) \mapsto y_{i_1}(t_1) \cdots y_{i_\ell}(t_\ell)B/B.$$

Proof. Let us define

$$xB \xrightarrow{w} zB \iff z \in xBwB.$$

For any $z \in G$, we have birational map

$$\mathbb{C}^\times \longrightarrow \{xB : zB \xrightarrow{s_i} xB\}, \quad t \longmapsto zy_i(t)B.$$

This can be seen from the SL_2 case. Then the Lemma follows from induction. $\square$

Let us define

$$g = y_{i_1}(t_1) \cdots y_{i_\ell}(t_\ell) \in BwB \cap U^-.$$

Since $gB \in BwB/B$, we have

$$g\eta_\lambda = \Delta_\lambda^{w\lambda}(g)\eta_{w\lambda} + (\text{vectors of weight } >_{\text{dom}} \lambda).$$

Actually the coefficient can be computed by induction.

Lemma. The minor

$$\Delta_\lambda^{w\lambda}(g) = \prod_{r=1}^{\ell} t_r^{\langle \alpha_{i_r}^\vee, w_{>r}\lambda \rangle}.$$

Proof. We have

$$\begin{aligned} g_{>1}\eta_\lambda &= \Delta_\lambda^{w_{>1}\lambda}(g_{>1})\eta_{w_{>1}\lambda} + (\text{higher}) \\ \Rightarrow g\eta_\lambda &= y_{i_1}(t_1)g_{>1}\eta_\lambda = y_{i_1}(t_1) \left(\Delta_\lambda^{w_{>1}\lambda}(g_{>1})\eta_{w_{>1}\lambda} + (\text{higher}) \right) \\ &= \Delta_\lambda^{w\lambda}(g)\eta_{w\lambda} + (\text{higher}). \end{aligned}$$

We see

$$\Delta_\lambda^{w\lambda}(g) = (\cdots) \Delta_\lambda^{w_{>1}\lambda}(g_{>1})$$

where $(\cdots)$ is the coefficient of

$$y_{i_1}(t_1)\eta_{w_{>1}\lambda} = (\cdots)\eta_\lambda + (\text{higher}).$$

Let us restricting the representation V^λ from G to $SL_2(\alpha_{i_1})$. Then $\eta_{w_{>1}\lambda}$ is of highest weight $\langle \alpha_{i_1}^\vee, w_{>1}\lambda \rangle$. By induction, it reduces to SL_2 case (computed below). $\square$

Example. For $G = SL_2$ and $\lambda = n$. We have realization

$$V^\lambda = \mathbb{C}[x, y]_{\deg=d}, \quad \eta_\lambda = x^d.$$

The action of SL_2 is by changing of variables. For example,

$$\begin{aligned} \dot{s} \cdot f(x, y) &= f\left(\begin{bmatrix} x & y \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\right) = f(y, -x). \\ y(t) \cdot f(x, y) &= f\left(\begin{bmatrix} x & y \\ t & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}\right) = f(x + ty, y). \end{aligned}$$

As a result,

$$\begin{aligned} \dot{s}_i \cdot x^d &= y^d \\ y(t) \cdot x^d &= (x + ty)^d = t^d y^d + \dots \end{aligned}$$

Since $gB \in BwB$, we can assume

$$gB = zwB, \quad z \in U \cap wU^-w^{-1} \subseteq U.$$

Since $z \in U$, we have

$$\Delta_{w\lambda}^\lambda(z) = \frac{\Delta_\lambda^\lambda(z\dot{w})}{1} = \frac{\Delta_\lambda^\lambda(z\dot{w})}{\Delta_\lambda^{w\lambda}(z\dot{w})} = \frac{\Delta_\lambda^\lambda(g)}{\Delta_\lambda^{w\lambda}(g)} = \frac{1}{\Delta_\lambda^{w\lambda}(g)}.$$

Lemma. We have

$$g_{\leq k}B = zw_{\leq k}B.$$

Proof. By construction,

$$B \xrightarrow{s_{i_1}} g_{\leq 1}B \xrightarrow{s_{i_2}} \dots \xrightarrow{s_{i_{\ell-1}}} g_{\leq \ell-1}B \xrightarrow{s_{i_\ell}} gB.$$

We have another chain

$$B = zB \xrightarrow{s_{i_1}} zw_{\leq 1}B \xrightarrow{s_{i_2}} \dots \xrightarrow{s_{i_{\ell-1}}} zw_{\leq \ell-1}B \xrightarrow{s_{i_\ell}} zwB.$$

Since the word is reduced, the two chains must be equal. $\square$

By the Lemma above, the same formula is true with w, g replaced by $w^{\leq k}$ and $g^{\leq k}$. We thus get the following.

Lemma. We have

$$\Delta_{w^{\leq k}\lambda}^\lambda(z) = \frac{1}{\Delta_\lambda^{w^{\leq k}\lambda}(g^{\leq k})} = \prod_{r=1}^k t_r^{-\langle \alpha_{i_r}^\vee, w_{>r}^{\leq k}\lambda \rangle}.$$

Chamber ansatz. We have

$$t_k = \frac{\prod_{j \neq i_k} \Delta_{w \leq k-1}^{\varpi_j}(z)^{-\langle \alpha_{i_k}, \alpha_j^\vee \rangle}}{\Delta_{w \leq k}^{\varpi_{i_k}}(z) \Delta_{w \leq k-1}^{\varpi_{i_k}}(z)}.$$

Proof. Note that

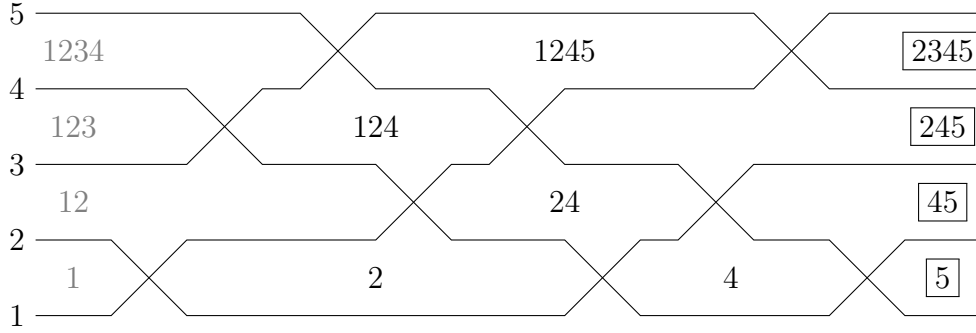
factor	power of t_k	power of other t_r
$\Delta_{\varpi_{i_k}}^{w \leq k}(g^{\leq k})$	1	$\langle w_{>r}^{\leq k}(\varpi_{i_k}), \alpha_{i_r}^\vee \rangle = \langle w_{>r}^{\leq k-1}(\varpi_{i_k} - \alpha_{i_k}), \alpha_{i_r}^\vee \rangle$
$\Delta_\lambda^{w \leq k-1}(g^{\leq k-1})$	0	$\langle w_{>r}^{\leq k-1}(\lambda), \alpha_{i_r}^\vee \rangle$

Using

$$\varpi_{i_k} - \alpha_{i_k} = -\varpi_{i_k} - \sum_{j \neq i} \langle \alpha_j^\vee, \alpha_{i_k} \rangle \varpi_j.$$

We can cancel the other t_r 's. □

Example. In type A , this formula admits a wire diagram explanation. Note that the meanings are different from cluster variables. The subset shows $w_{\leq k} \varpi_{i_k}$.



By above computation,

$$t_k = \frac{UD}{LR}, \quad \begin{array}{c} U \\ \overline{L \quad R} \\ D \end{array}$$

5.2. Twist map. The computation of this part is very formal.

Let us denote the Chevalley involution by θ . It sends

$$\text{image of } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \text{image of } \begin{bmatrix} d & c \\ b & a \end{bmatrix}.$$

We thus get $\theta(\dot{w}) = \ddot{w}$.

Lemma. We have

$$\Delta_{u\lambda}^{v\lambda}(\theta(g)) = \Delta_{-u\lambda}^{-v\lambda}(g).$$

Proof. Let us first prove for $u = v = e$. It suffices to check the identity over the dense subset $B^+B^- = U^+TU^-$. Write $g = u_1tu_2$. Then

$$\begin{aligned}\Delta_\lambda^\lambda(\theta(g)) &= \Delta_\lambda^\lambda(\theta(u_1)\theta(t)\theta(u_2)) = \Delta_\lambda^\lambda(\theta(t)) = t^{-\lambda} \\ \Delta_{-\lambda}^{-\lambda}(g) &= \Delta_{-\lambda}^{-\lambda}(u_1tu_2) = \Delta_{-\lambda}^{-\lambda}(t) = t^{-\lambda}.\end{aligned}$$

Here we use $-\lambda$ is antidominant, $u_1, \theta(u_2) \in U^+$ and $u_2, \theta(u_1) \in U^-$.

In general, we have

$$\Delta_{u\lambda}^{v\lambda}(\theta(g)) = \Delta_\lambda^\lambda(\dot{v}^{-1}\theta(g)\dot{u}) = \Delta_{-\lambda}^{-\lambda}(\ddot{v}^{-1}g\ddot{u}) = \Delta_{-v\lambda}^{-u\lambda}(g)$$

Note that $\eta_{-w\lambda} = \ddot{w}\eta_{-\lambda}$. □

Lemma. We have

$$\Delta_\lambda^\lambda(g) = \Delta_{-\lambda}^{-\lambda}(g^{-1}), \quad \Delta_{-\lambda}^{-\lambda}(g) = \Delta_\lambda^\lambda(g^{-1})$$

Proof. The proof is similar. □

Let us consider a twist map

$$\begin{array}{ccccccc} Bw^{-1}B/B & \xleftarrow{\sim} & U \cap w^{-1}U^-w & \xleftarrow{\sim} & U \cap wU^-w^{-1} & \xrightarrow{\sim} & BwB/B \\ yw^{-1}B/B & \xleftarrow{\sim} & y & \xrightarrow{\sim} & z & \xrightarrow{\sim} & zwB/B \end{array}$$

where

$$z = \dot{w}\theta(y)^{-1}\dot{w}^{-1}, \quad y = \ddot{w}^{-1}\theta(z)^{-1}\ddot{w}.$$

If we denote $y \mapsto z$ by f_w , then $f_{w^{-1}} \circ f_w = \text{id}$.

Lemma. Assume u, v appears as a suffix of w , we have

$$\begin{aligned}\Delta_{u\lambda}^{v\lambda}(z) &= \Delta_{u\lambda}^{v\lambda}(\dot{w}\theta(y)^{-1}\dot{w}^{-1}) \\ &= \Delta_\lambda^\lambda(\dot{v}^{-1}\dot{w}\theta(y)^{-1}\dot{w}^{-1}\dot{u}) \\ &= \Delta_{-\lambda}^{-\lambda}(\dot{u}^{-1}\dot{w}\theta(y)\dot{w}^{-1}\dot{v}) \\ &= \Delta_\lambda^\lambda(\ddot{u}^{-1}\ddot{w}y\ddot{w}^{-1}\ddot{v}) = \Delta_{w^{-1}v\lambda}^{w^{-1}u\lambda}(y).\end{aligned}$$

Corollary. In particular,

$$\Delta_{w \leq k \varpi_{i_k}}^{\varpi_{i_k}}(z) = \Delta_{w^{-1} \varpi_{i_k}}^{w^{-1} \varpi_{i_k}}(y) = \Delta_{\varpi_{i_k}}^{w^{-1} \varpi_{i_k}}(y\ddot{w}^{-1}).$$

That is, after composing the twist map, the minors $\Delta_{w \leq k \varpi_{i_k}}^{\varpi_{i_k}}(z)$ are exactly the cluster variables.

5.3. Proof of (1). By Chamber Ansats and the Lemma before Chamber Ansats, we see $\{\Delta_{w \leq^k \varpi_i}^{\varpi_i}(z)\}$ and $\{t_1, \dots, t_\ell\}$ represent each other by Laurent monomials. In particular, the image of

$$(\mathbb{C}^\times)^\ell \longrightarrow BwB/B$$

is exactly

$$\{zwB \in BwB/B : \Delta_{w \leq^k \varpi_{i_k}}^{\varpi_{i_k}}(z) \neq 0\}.$$

After composing with the twist map, we get the proof of (1).

6. PROOF OF (3)

The argument of this section is from [3]. Note that the map in (3) is always well-defined. It suffices to construct the inverse map. The the inverse map is actually known, as a rational function, given by

$$(\mathbb{C}^\times)^\ell \xrightarrow{(*)} BwB/B \xrightarrow{\sim} Bw^{-1}B/B.$$

Here $(*)$ is given by sending

$$\begin{aligned} (\Delta'_i)_{i=1}^\ell &\longmapsto (\Delta_i)_{i=1}^\ell && \text{use the mutation at } k \\ &\longmapsto (t_i)_{i=1}^\ell && \text{use Chamber Ansats} \\ &\longmapsto g = y_{i_1}(t_1) \cdots y_{i_\ell}(t_\ell) \\ &\longmapsto gB. \end{aligned}$$

That is, $\Delta'_i = \Delta_i$ if $i \neq k$ and

$$\Delta_k \Delta'_k = \text{a bimonomial in } \Delta_i \text{ as in the mutation,}$$

and

$$t_i = \text{a Laurent monomial in } \Delta_i \text{ as in Chamber Ansats.}$$

If we can show $(*)$ is actually a regular map (instead of a rational map), then (3) is proved. Actually the image in $Bw^{-1}B/B$ is automatically contained in $\overset{\circ}{R}_{e,w^{-1}}$, as the frozen variables $\Delta_{\varpi_i}^{\varpi_i}$ are unchanged and thus nonzero.

The strategy is show

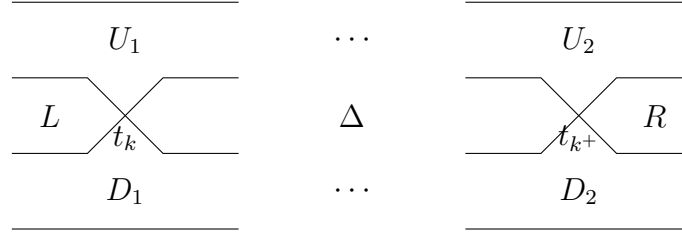
- (1) $(\Delta'_i)_i \longmapsto g \in G$ is a regular map,
- (2) and its image is in BwB .

We can prove (2) if we know (1). Firstly, the image has to be in the closure $\overline{BwB}$. Assume $(\Delta'_i)_i \mapsto g \in G$, then

$$\Delta_{\varpi_i}^{w\varpi_i}(g) = 1.$$

This is the trivial case of the Chamber Ansats. Thus the image is actually in BwB .

6.1. **Regularity.** Assume the k -th minor corresponds to a closed chamber at the i -th level:



We can view $t_1, \dots, t_\ell$ as a rational function in (Δ'_i) . Since mutation does not change t_r for $r \notin [k, k^+]$, it suffices to show for each mutable k

$$y_{i_k}(t_k) \cdots y_{i_{k^+}}(t_{k^+}) \text{ is regular.}$$

It suffices to show

$$(\dagger) \quad y_i(t_k)y_{i_m}(t_m)y_i(-t_k), \quad k < m < k^+,$$

$$(\ddagger) \quad y_i(t_k)y_{i_{k^+}}(t_{k^+})$$

are all regular, since the product of them is exactly the RHS.

We have

$$y_i(t_k)y_{i_{k^+}}(t_{k^+}) = y_i(t_k + t_{k^+}).$$

When each $j-i$ appears in the interval $i_k, \dots, i_{k^+}$.

$$t_k + t_{k^+} = \frac{U_1 D_1}{L \Delta} + \frac{U_2 D_2}{\Delta R} = \frac{U_1 D_1 R + L U_2 D_2}{L \Delta R} = \Delta' \frac{1}{LR}.$$

In general, if $j = i - 1$ does not appear and $j = i + 1$ appears, then $D_1 = D_2$, and

$$t_k + t_{k^+} = \frac{U_1 D_1}{L \Delta} + \frac{U_2 D_2}{\Delta R} = D_1 \frac{U_1 R + L U_2}{L \Delta R} = \Delta' \frac{D_1}{LR}.$$

Thus $(\ddagger)$ is regular.

Let us prove $(\dagger)$ is regular. When i_m is not connected to i , mutation does not change t_m , and

$$y_i(t_k)y_{i_m}(t_m)t_i(-t_k) = y_{i_m}(t_m)$$

which is regular. When $i_m - i$, by Lemma below we have

$$y_i(t_k)y_{i_m}(t_m)t_i(-t_k)$$

is regular if t_m and $t_k t_m$ are both regular. Note that

$$\begin{aligned} t_m &= \Delta \cdot (\text{a Laurent monomial in other}) \\ &= \frac{\text{a binomial}}{\Delta'} \cdot (\text{a Laurent monomial}) \\ &= \text{a Laurent monomial monomial in } \{\Delta'_i\} \\ t_k t_m &= \frac{U_1 D_1}{L \Delta} \Delta \cdot (\text{a Laurent monomial in other}) \\ &= (\text{a Laurent monomial in other}). \end{aligned}$$

Both are regular functions, so we are done.

Lemma. If $i \rightarrow j$, then

$$y_i(x)y_j(y/x)y_i(x)$$

can be extended to $x, y \in \mathbb{C}$.

Proof. It suffices to show for SL_3 . We have

$$\begin{bmatrix} 1 & & \\ x & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ y/x & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ -x & 1 & \\ & & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ -y & 1 & \\ & y & 1 \end{bmatrix}$$

□

7. DEPENDENCE OF REDUCED WORDS

Plücker Relation ([2, Theorem 1.6]). If $\langle \alpha_i, \alpha_j \rangle = -1$, $vs_i > v$, $vs_j > v$, then

$$\Delta_{u\varpi_i}^{vs_i\varpi_i} \Delta_{u\varpi_j}^{vs_j\varpi_j} = \Delta_{u\varpi_i}^{v\varpi_i} \Delta_{u\varpi_j}^{vs_i s_j \varpi_j} + \Delta_{u\varpi_i}^{vs_j s_i \varpi_i} \Delta_{u\varpi_j}^{v\varpi_j}.$$

Proof. It suffices to check when $u = v = \text{id}$. It reduces to check SL_3 . Assume

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}$$

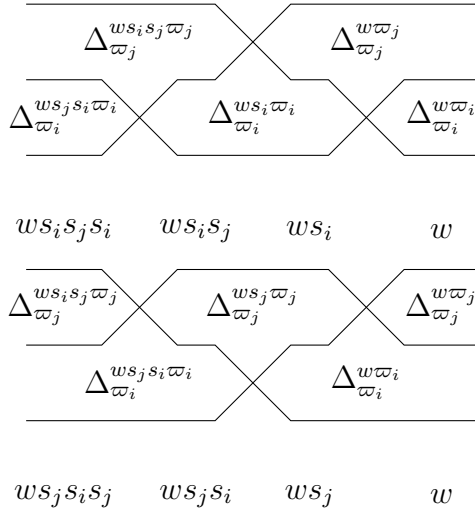
this equality says

$$x_{12} \begin{vmatrix} x_{11} & x_{13} \\ x_{21} & x_{23} \end{vmatrix} = x_{11} \begin{vmatrix} x_{12} & x_{13} \\ x_{22} & x_{23} \end{vmatrix} + x_{13} \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}.$$

It follows from $\text{RHS} - \text{LHS} = \begin{vmatrix} x_{11} & x_{12} & x_{13} \\ x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{vmatrix} = 0.$

□

This is exactly the mutation relation:

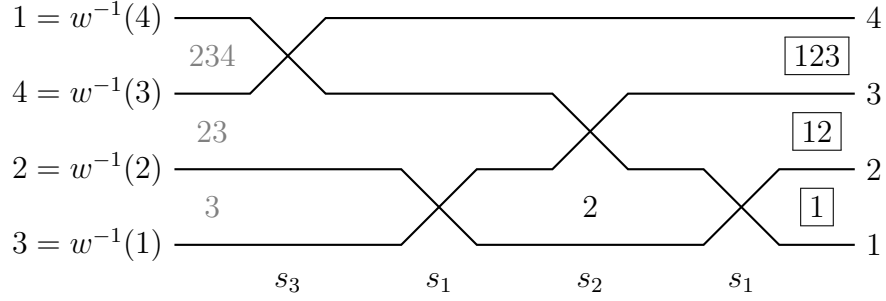


8. EXAMPLE

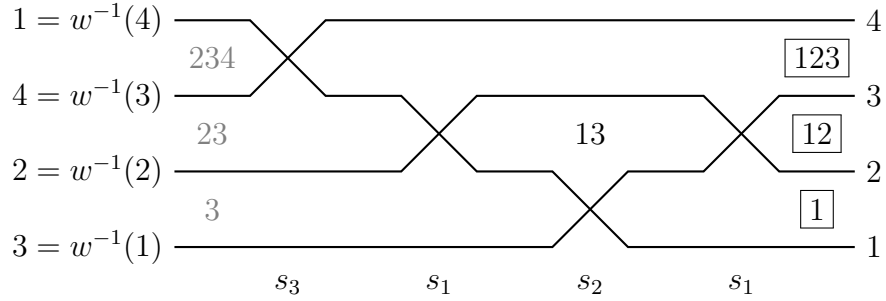
Let $w = s_3 s_1 s_2 s_1$. We have

$$\begin{array}{ccccc}
 w^{-1} & w_{>1}^{-1} = s_1 s_2 s_1 & w_{>2}^{-1} = s_1 s_2 & w_{>3}^{-1} = s_1 & w_{>4}^{-1} \\
 234 & 123 & 123 & 123 & 123 \\
 23 & 23 & 23 & 12 & 12 \\
 3 & 3 & 2 & 2 & 1
 \end{array}$$

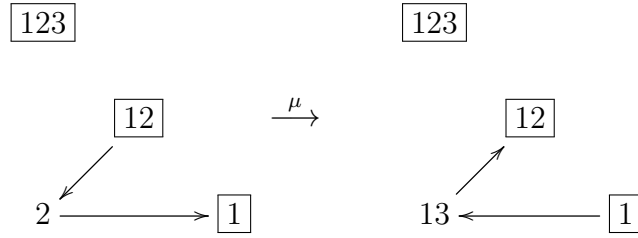
We construct a wiring diagram



Another reduced word is $w = s_3 s_2 s_1 s_2$.



The quiver is



We have

$$\ddot{w}^{-1} = \dot{s}_2 \dot{s}_1 \dot{s}_2 \dot{s}_3 = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and

$$\ddot{w}^{-1}U \cap U^{-}\ddot{w}^{-1} = \left\{ \begin{bmatrix} x & y & z & -1 \\ w & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} : x, y, z, w \in \mathbb{C} \right\}.$$

We have

	subsets	minor		subsets	minor
	subsets	minor	23	$\begin{vmatrix} w & -1 \\ 1 & 0 \end{vmatrix} = 1$	
3	1			234	$\begin{vmatrix} w & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$
2	w		13	$\begin{vmatrix} x & y \\ 1 & 0 \end{vmatrix} = -y$	
1	x			123	$\begin{vmatrix} x & y & z \\ w & -1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = z$
			12	$\begin{vmatrix} x & y \\ w & -1 \end{vmatrix} = -x - wy$	

We do have

$$A_2 A_{13} = w \cdot (-y) = (-x - wy) + x = A_{12} + A_1.$$

The cluster algebra

$$\begin{aligned} & \mathbb{C}[A_{123}, A_{12}, A_1, A_2, A_{13}] / (A_2 A_{13} = A_{12} + A_1) \\ & \cong \mathbb{C}[A_{123}, A_1, A_2, A_{13}] = \mathbb{C}[z, x, w, -y]. \end{aligned}$$

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